Implementation and Evaluation of "An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning"

CS 552 Group Project — Pratham Garg, Sumit Sharma, Vivaan Garg

Assigned paper. S. Krishnan and S. M. Naik, *Security and Privacy* 8(4): e70057, 2025. DOI 10.1002/spy2.70057.

1 Task and approach

The task is to implement the methodology of the paper and reproduce its experiments as closely as possible. The paper detects phishing from the URL string alone and keeps the detector current under concept drift. A one-time offline phase selects features with LASSO and mutual information (MI) and trains an SGD classifier. In the online phase a Kolmogorov–Smirnov (KS) monitor watches each feature on incoming mini-batches. When the distribution shifts, the feature set is re-selected and the classifier is updated incrementally, without full retraining.

2 Implementation

All components are implemented from the paper's description in the Python package `drift_phish`.

- **Features (Table 3).** 58 lexical features from the URL string only (lengths, token and character counts, delimiters, digits, entropy, scheme, IP-in-host, suspicious keywords, obfuscation indicators). No HTML is fetched and no third-party service is queried; redirect-chain and domain-mismatch indicators are lexical proxies.
- **Dynamic feature selection (Alg. 1).** Lasso ($\lambda = 10^{-4}$, 1000 iterations) gives weights w; MI gives scores I; $S = w + I$; the selected set is $F^* = \{S \geq \tau\}$ with τ the 75th percentile. On drift, scores are blended with the importance history H; drifted features are marked obsolete and replaced.
- **Drift monitor.** Per-feature two-sample KS test, $\delta = 0.05$, reference buffer of 1500, mini-batches $B = 500$. A drift event is declared when at least 25% of the features in F^* fail the test (the paper does not specify the aggregation rule).
- **Online learner.** SGD with hinge (SGD-SVM), log (SGD-LR) or perceptron loss; adaptive learning rate, $\eta_0 = 0.01$, balanced class weights, early stopping, seed 42. The classifier lives in the full feature space and F^* acts as a mask, so learned weights survive feature changes; after re-selection a few replay epochs refresh the model.
- **Evaluation.** 80:20 stratified split for the offline phase; prequential (test-then-train) evaluation on the stream. Positive class = phishing.

3 Data and protocol

- **PhiUSIIL** (UCI 967): 235k URLs. **Kaggle** "Phishing Site URLs": 21k balanced subset. **ISCX-URL-2016**: available as the authors' 79 pre-computed lexical features (15,367 benign/phishing rows, no URL strings); used for Table 5. **Hannousse & Yahiouhmed 2021**: 11,430 URLs, used where raw URL strings are needed.
- **Combined corpus:** PhiUSIIL + Kaggle + Hannousse, de-duplicated (263k URLs).
- **New URLs:** live PhishTank verified-online feed (snapshot 2026-11-06), restricted to the paper's collection window (June 2024 – January 2025): 12,888 phishing URLs not present in the old corpora, streamed in chronological order with an equal number of held-out benign URLs.

4 Results

4.1 Offline phase (paper Tables 5–7, Fig. 6)

SGD-SVM is the strongest of the three SGD variants on Kaggle, ISCX and the combined corpus, and within 0.01 point of the best on PhiUSIIL, as in the paper. On PhiUSIIL the result matches the paper. On the other

Corpus	Source	Acc.	Prec.	Rec.	F1
PhiUSIIL	this study	99.38	99.99	98.54	99.26
	paper (Table 7)	98.78	97.84	98.73	98.28
Kaggle	this study	73.81	81.84	60.33	69.46
	paper (Table 6)	99.00	98.92	98.77	98.85
ISCX	this study (79 features)	90.83	90.49	90.97	90.73
	paper (Table 5)	99.02	98.39	99.63	99.01
Combined	this study	93.58	91.78	93.33	92.55
	paper (Fig. 6)	99.85	—	—	—

Table 1: SGD-SVM on LASSO+MI features, 80:20 split, seed 42 (%). FPR of this study: 0.004% (PhiUSIIL), 13.0% (Kaggle), 9.3% (ISCX), 6.2% (combined); paper's combined FPR: 0.067%.

corpora the absolute level is lower: using all ISCX features instead of the 20 selected raises SGD-SVM accuracy from 90.8% to 96.1%, and a random forest on the same 20 features reaches 98.3%, so the level depends on the feature set and corpus version in use.

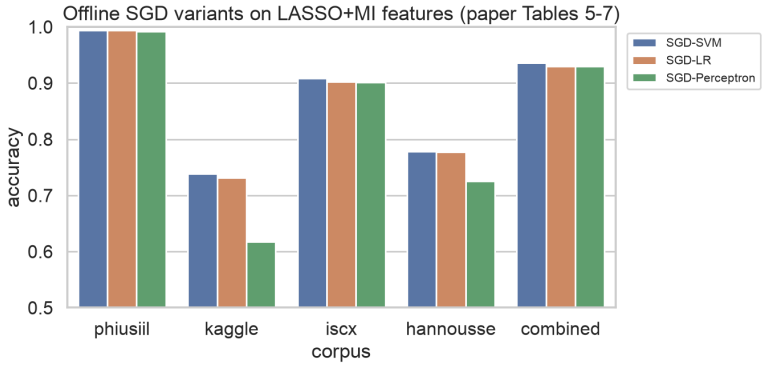


Figure 1: Offline SGD variants per corpus (cf. paper Fig. 6).

4.2 Drift and feature evolution (paper Figs. 4, 5, 7, Table 8)

Between 1500 old phishing URLs and 500 new PhishTank URLs, the KS test at $\delta = 0.05$ flags 34 of 58 features. The largest shifts are in host entropy, longest host label, domain length, scheme (`https`/`http`), path length and URL length, i.e. the same length- and structure-type features the paper reports as drifting. Feature sets after successive drift events are listed in `RESULTS.md` (Table 8 analogue): for example drift stage 1 on the PhishTank stream adds `entropy`, `host_entropy` and `longest_host_label_length` and removes `has_https`, `num_dots` and `scheme_http`.

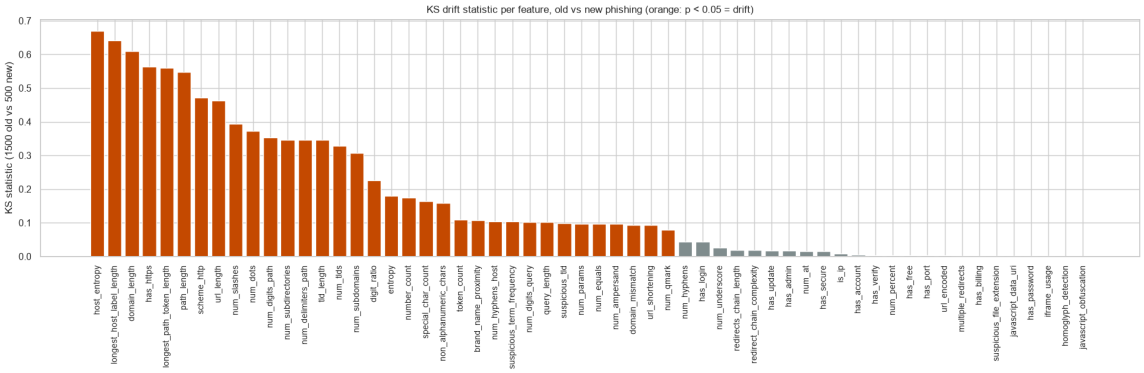


Figure 2: KS statistic per feature, old vs. new phishing URLs (cf. paper Fig. 7).

System	Acc.	F1	FPR	Drift events
Frozen offline model	92.63 ± 6.43	92.03	6.25	0
Online SGD, static F^*	97.77 ± 0.16	97.79	3.10	0
Proposed (KS + adaptive F^* + online SGD)	**97.80 ± 0.05**	**97.82**	3.16	28.6
Periodic batch retraining	97.31 ± 0.61	97.33	3.69	5

Table 2: Prequential results on the 25,776-URL PhishTank stream, 5 seeds (mean ± std, %).

4.3 Online phase

Online learning improves on the frozen offline model by 5.1 points of accuracy, with far lower variance across seeds. Adaptive feature selection is at least as good as the static feature set. In a stronger cross-corpus shift (offline PhiUSIIL, then Kaggle, then Hannousse) the frozen model drops to 55.5%, online SGD with a static feature set reaches 70.5% and the proposed framework 73.2%.

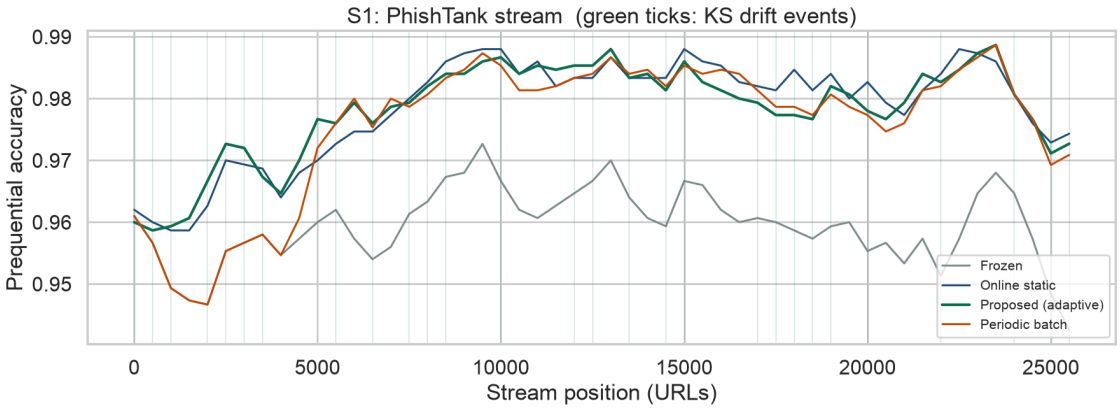


Figure 3: Prequential accuracy over the PhishTank stream (cf. paper Fig. 8).

4.4 Effect of adaptive feature selection (paper Table 9)

	100 URLs		500 URLs		1000 URLs	
	static	adaptive	static	adaptive	static	adaptive
This study (F1, %)	96.19	96.08	96.25	96.90	96.54	96.85
Paper (F1, %)	92.31	96.22	93.89	97.15	93.73	98.30

Table 3: F1 on held-out new URLs after adapting with N labelled new URLs (10 seeds).

The F1 of the adaptive framework is within 0.1–1.5 points of the paper's values. Our static baseline is stronger than the paper's, so the gap between static and adaptive is smaller; at $N \geq 500$ adaptive selection still improves accuracy (+0.7 point) and lowers the false-positive rate (6.8% to 5.4%). Under a larger shift (offline PhiUSIIL, target Hannousse) adaptive selection improves accuracy by up to 12 points.

4.5 Runtime (paper Tables 10–11)

Feature extraction plus scoring takes 0.16 ms per URL on one CPU thread (paper: 12.4–15.2 ms; different hardware and implementation). An incremental update on 100–1000 URLs takes 1.5–1.9 ms versus about 4 s for batch retraining on the 210k-URL training set (more than 2000×); adapting the feature set costs 213 ms versus 2.7 s for full retraining (12.9×). The paper reports gains of 15.9–33.7×; the conclusion that incremental updating is far cheaper than retraining is the same.

5 Evaluation against the assignment

Requirement / paper claim	Status	Evidence
Implement the proposed methodology	✓	Features, LASSO+MI, KS monitor, online SGD, full pipeline
Source code	✓	`Source_Code/` (16 unit tests passing)
Implementation and results	✓	`Experimental_Results/`
Short project report	✓	this document
Presentation and demo	✓	slides; browser demo in `Source_Code/web`
Full title and complete reference on slide 1	✓	slide 1
SGD-SVM is the best SGD variant	✓	Tables above
PhiUSIIL accuracy (Table 7)	✓	99.38% vs. 98.78%
Feature distributions drift between old and new URLs	✓	34/58 features, same feature types
Feature set evolves after drift (Table 8)	✓	per-event additions and removals logged
Online learning beats a frozen model	✓	+5.1 points (stream), +15.0 (cross-corpus)
Adaptive features, F1 at $N = 100/500/1000$ (Table 9)	✓	96.08 / 96.90 / 96.85 vs. paper 96.22 / 97.15 / 98.30
Incremental updates cheaper than retraining (Table 11)	✓	>2000× (update), 12.9× (features)
Real-time per-URL latency (Table 10)	✓	0.16 ms per URL
Kaggle / ISCX / combined accuracy ≈99% (Tables 5, 6, Fig. 6)	∼	73.8% / 90.8% / 93.6%; see Section 6

Table 4: ✓ = consistent with the paper; ∼ = same ranking of methods but lower absolute level.

6 Discussion

All components of the method were implemented and every experiment of the paper's results section was run. The qualitative findings — SGD-SVM is the best of the three linear learners, features such as URL, path and host length drift between old and new phishing URLs, online learning with drift-triggered feature re-selection holds accuracy where a frozen model degrades, and incremental updates cost a small fraction of retraining — are all observed. PhiUSIIL accuracy and the adaptive F1 values of Table 9 are close to the paper.

Absolute accuracies on the Kaggle, ISCX and combined corpora are lower than in the paper. The paper does not publish its exact corpus versions, URL pre-processing or full feature definitions, and the results here show how much these matter: the same learner moves from 73.8% to 79.8% on Kaggle and from 90.8% to 96.1% on ISCX simply by using all features rather than the selected quarter, and tree ensembles on the same features reach 86% (Kaggle) and 98% (ISCX). Our PhiUSIIL benign URLs are mostly bare domains, so a model trained on the combined corpus separates PhiUSIIL almost perfectly (98.5%) but is much weaker on the other two corpora, which lowers the combined figure. The benign side of the PhishTank stream is held-out old benign URLs because PhishTank itself lists only phishing.

Limitations. Lexical features cannot detect phishing that has no suspicious URL pattern. The 25% drift-aggregation rule, history weight and replay epochs are our choices where the paper is silent. Absolute numbers depend on the corpus versions listed above, and a live feed changes daily, so the PhishTank snapshot is stored with the code.

7 Reproducing

`cd Source_Code && pip install -r requirements.txt && make test && make all` downloads the data, runs experiments 1–5 and regenerates `Experimental_Results/RESULTS.md` and all figures.